

Cheese Shelf-Life Studio — V7 External Test Evaluation

21 real, literature-sourced cases from CHEESE_SHELF_LIFE_V7_EXTERNAL_TEST_POINT_TARGETS.csv, scored against the live V7 specialist pipeline. Generated 2026-08-24 22:55 UTC. Raw model output only — no correction, multiplier, or manual calibration applied.

Executive summary

- All 21 rows produced a prediction (none hard-blocked) — but **all 21 were flagged unsupported_categorical** : every case uses at least one food_matrix / packaging_type / indicator_type / physical_form value absent from V7's training vocabulary. See §3.
- Overall: n=21, MAE=39.4d, RMSE=65.8d, bias=-30.0d, R²=0.172.
- Semi-hard** generalizes best (R²=0.560); **Hard** has effectively collapsed on real data (R²=-3.112, bias=-111.9d) — see §5.
- A real bug in the support-assessment logic (missing numeric/categorical values misreported as false extrapolation/unseen-category flags) was found and fixed while building this evaluation — see §6.
- No food_matrix/physical_form/packaging_type/indicator_type value was remapped or guessed to improve the score — see §2.

1. Test set

21 cases across 6 independent papers, all with exact ("point", zero-censoring) targets and direct_model_compatible=yes . All rows are general_shelf_life (no safety-endpoint cases in this file). Category split: soft n=9, semi-hard n=7, hard n=5.

2. Methodology — what was and wasn't normalized

Evaluation uses the exact production code path: SpecialistRegistry.resolve() / assess_prediction_support() / ModelService.predict_one() — the same functions /api/v6/predict calls, not a separate reimplementation. Two, and only two, normalizations were applied, both pure notation (verified, not guessed):

- cheese_category spelling: "semi-hard" → "semi_hard" . This is a routing key only — cheese_category is not even a model feature.
- Concentration units run through the same to_canonical() a live request would use; where the file's unit isn't in the verified table (%_coating_solution , 2 rows), the conversion is left unavailable (None), not guessed.

Nothing else was remapped. food_matrix, physical_form, packaging_type, and indicator_type values that don't match V7's training vocabulary are passed through as-is and flagged by the existing support-assessment machinery, exactly as a real live prediction would be.

3. Aggregate results

SLICE	N	MAE (D)	RMSE (D)	BIAS (D)	R ²
Overall	21	39.4	65.8	-30.0	0.172
Soft	9	12.4	20.4	-7.5	-0.120
Semi-hard	7	22.3	25.8	-0.4	0.560
Hard	5	111.9	128.6	-111.9	-3.112

By support level (all 21 predicted rows fall in one bucket — see below):

SUPPORT LEVEL	N	MAE (D)	RMSE (D)	BIAS (D)	R ²
unsupported categorical	21	39.4	65.8	-30.0	0.172

Every one of the 21 predictable rows was flagged unsupported_categorical — there is no "clean" supported subset to compare against in this file. This is the hardened pipeline behaving as designed (it never silently presents an out-of-vocabulary input as a normal-confidence result), but it means the R²/MAE above conflate genuine model error with vocabulary-coverage gaps. Treat this file as a coverage stress test more than a clean accuracy benchmark until a version using V7's own vocabulary is available.

4. Full per-row detail

★ marks the specialist's production model (lowest validation RMSE). Every flag is listed in full — nothing truncated.

EXT_ALVES96_AIR — soft / Mozzarella

true 13d → pred 10.9d (err -2.1d)

Alves et al. (1996), Shelf Life of Sliced Mozzarella Cheese in Modified-Atmosphere Packaging · DOI 10.4315/0362-028X-59.8.838 · 1996

Sliced mozzarella stored at 7±1°C under the reported package atmosphere.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
4.5	10.9 ★	9.4	19.2	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Mozzarella

unseen packaging_type=air_PVC

unseen indicator_group=sensory_quality

unseen indicator_type=overall_sensory_deterioration

unseen treatment_type=conventional_packaging

unseen application_method=packaging_atmosphere

unseen physical_form=sliced

'headspace_oxygen_pct' = 21.0 is outside the training range [0.5, 20.9]

'headspace_co2_pct' = 0.0 is outside the training range [0.04, 50]

EXT_ALVES96_CO2 — soft / Mozzarella

true 63d → pred 12.5d (err -50.5d)

Alves et al. (1996), Shelf Life of Sliced Mozzarella Cheese in Modified-Atmosphere Packaging · DOI 10.4315/0362-028X-59.8.838 · 1996

Sliced mozzarella stored at 7±1°C under the reported package atmosphere.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
4.8	12.5 ★	10.7	23.4	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Mozzarella

unseen packaging_type=MAP_100_CO2

unseen indicator_group=sensory_quality

unseen indicator_type=overall_sensory_deterioration

unseen treatment_type=modified_atmosphere_packaging

unseen application_method=packaging_atmosphere

unseen physical_form=sliced

'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]

'headspace_co2_pct' = 100.0 is outside the training range [0.04, 50]

EXT_ALVES96_CO2N2 — soft / Mozzarella

true 45d → pred 12.4d (err -32.6d)

Alves et al. (1996), Shelf Life of Sliced Mozzarella Cheese in Modified-Atmosphere Packaging · DOI 10.4315/0362-028X-59.8.838 · 1996

Sliced mozzarella stored at 7±1°C under the reported package atmosphere.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
4.8	12.4 ★	10.7	22.3	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Mozzarella

unseen packaging_type=MAP_50_CO2_50_N2

unseen indicator_group=sensory_quality

unseen indicator_type=overall_sensory_deterioration

unseen treatment_type=modified_atmosphere_packaging

unseen application_method=packaging_atmosphere

unseen physical_form=sliced

'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]

EXT_ALVES96_N2 — soft / Mozzarella

true 16d → pred 11.8d (err -4.2d)

Alves et al. (1996), Shelf Life of Sliced Mozzarella Cheese in Modified-Atmosphere Packaging · DOI 10.4315/0362-028X-59.8.838 · 1996

Sliced mozzarella stored at 7±1°C under the reported package atmosphere.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
4.5	11.8 ★	10.3	21.0	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Mozzarella

unseen packaging_type=MAP_100_N2

unseen indicator_group=sensory_quality

unseen indicator_type=overall_sensory_deterioration

unseen treatment_type=modified_atmosphere_packaging

unseen application_method=packaging_atmosphere

unseen physical_form=sliced

'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]

'headspace_co2_pct' = 0.0 is outside the training range [0.04, 50]

'headspace_n2_pct' = 100.0 is outside the training range [0, 79.1]

EXT_AZHDARI22_CMC — soft / High-moisture Mozzarella

true 8d → pred 11.2d (err +3.2d)

Azhdari & Moradi (2022), CMC/natamycin coating of high-moisture mozzarella cheese · DOI 10.1016/j.ijbiomac.2022.04.185 · 2022

High-moisture mozzarella at 7°C with the reported coating treatment.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
4.5	11.2 ★	10.4	21.3	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=High-moisture Mozzarella unseen indicator_group=total_bacterial_count unseen indicator_type=total_mesophilic_count
unseen indicator_unit=log10_CFU_g unseen application_method=immersion_coating unseen primary_ingredient_name=carboxymethyl_cellulose
unseen primary_ingredient_family=hydrocolloid_coating unseen physical_form=unknown

EXT_AZHDARI22_CTRL — soft / High-moisture Mozzarella

true 4d → pred 10.9d (err +6.9d)

Azhdari & Moradi (2022), CMC/natamycin coating of high-moisture mozzarella cheese · DOI 10.1016/j.ijbiomac.2022.04.185 · 2022

High-moisture mozzarella at 7°C with the reported coating treatment.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
4.5	10.9 ★	9.5	19.1	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=High-moisture Mozzarella unseen indicator_group=total_bacterial_count unseen indicator_type=total_mesophilic_count
unseen indicator_unit=log10_CFU_g unseen treatment_type=control unseen physical_form=unknown

EXT_AZHDARI22_NAT005 — soft / High-moisture Mozzarella

true 8d → pred 11.7d (err +3.7d)

Azhdari & Moradi (2022), CMC/natamycin coating of high-moisture mozzarella cheese · DOI 10.1016/j.ijbiomac.2022.04.185 · 2022

High-moisture mozzarella at 7°C with the reported coating treatment.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
4.6	11.7 ★	10.5	22.3	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=High-moisture Mozzarella unseen indicator_group=total_bacterial_count unseen indicator_type=total_mesophilic_count
unseen indicator_unit=log10_CFU_g unseen application_method=immersion_coating unseen physical_form=unknown
concentration unit unrecognized: No verified conversion for concentration unit '%_coating_solution'. Known units: ['%', '% (v/v)', '% (v/v)', '% w/v', '% w/w', '%(w/v)', '(w/v) %', '(w/w) %', 'AU/g', 'IU/g', 'g', 'g/L', 'g/kg', 'hr', 'log CFU/g', 'mg/5 ml', 'mg/L', 'mg/g', 'mg/kg', 'mg/mL', 'ml', 'mmol/L', 'none', 'ppm', 'wt %']

EXT_AZHDARI22_NAT05 — soft / High-moisture Mozzarella

true 8d → pred 11.7d (err +3.7d)

Azhdari & Moradi (2022), CMC/natamycin coating of high-moisture mozzarella cheese · DOI 10.1016/j.ijbiomac.2022.04.185 · 2022

High-moisture mozzarella at 7°C with the reported coating treatment.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
4.6	11.7 ★	10.5	22.3	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=High-moisture Mozzarella unseen indicator_group=total_bacterial_count unseen indicator_type=total_mesophilic_count
unseen indicator_unit=log10_CFU_g unseen application_method=immersion_coating unseen physical_form=unknown
concentration unit unrecognized: No verified conversion for concentration unit '%_coating_solution'. Known units: ['%', '% (v/v)', '% (v/v)', '% w/v', '% w/w', '%(w/v)', '(w/v) %', '(w/w) %', 'AU/g', 'IU/g', 'g', 'g/L', 'g/kg', 'hr', 'log CFU/g', 'mg/5 ml', 'mg/L', 'mg/g', 'mg/kg', 'mg/mL', 'ml', 'mmol/L', 'none', 'ppm', 'wt %']

EXT_BARUK20_AIR — soft / Fresh acid-coagulated cheese

true 9d → pred 13.7d (err +4.7d)

Barukčić et al. (2020), Effect of different packaging conditions on shelf life of fresh acid-coagulated cheese · DOI 10.1007/s13197-020-04308-6 · 2020

Fresh acid-coagulated cheese at 4±0.5°C in atmospheric air.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
5.6	13.7 ★	11.8	22.2	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Fresh acid-coagulated cheese unseen packaging_type=atmospheric_air unseen indicator_group=combined_quality
unseen indicator_type=microbiological_and_sensory_shelf_life unseen treatment_type=conventional_packaging unseen physical_form=unknown

EXT_GOUDA_SPREAD18_20C — semi_hard / Processed Gouda cheese spread

true 28d → pred 43.1d (err +15.1d)

Processed Gouda cheese spread shelf-life study (2018) · DOI 10.1007/s10068-018-0377-2 · 2018

Spreadable processed Gouda stored at 20°C.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
42.7	43.1 ★	40.2	0.0	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Processed Gouda cheese spread unseen indicator_group=sensory_quality unseen indicator_type=taste_smell_consistency_change
unseen treatment_type=temperature_storage unseen physical_form=spreadable 'storage_temperature_c' = 20.0 is outside the training range [2, 15]

EXT_GOUDA_SPREAD18_30C — semi_hard / Processed Gouda cheese spread

true 4d → pred 43.1d (err +39.1d)

Processed Gouda cheese spread shelf-life study (2018) · DOI 10.1007/s10068-018-0377-2 · 2018

Spreadable processed Gouda stored at 30°C.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
42.7	43.1 ★	40.2	0.0	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Processed Gouda cheese spread unseen indicator_group=sensory_quality unseen indicator_type=taste_smell_consistency_change
unseen treatment_type=temperature_storage unseen physical_form=spreadable 'storage_temperature_c' = 30.0 is outside the training range [2, 15]

EXT_GOUDA_SPREAD18_8C — semi_hard / Processed Gouda cheese spread

true 49d → pred 53.0d (err +4.0d)

Processed Gouda cheese spread shelf-life study (2018) · DOI 10.1007/s10068-018-0377-2 · 2018

Spreadable processed Gouda stored at 8°C.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
51.0	53.0 ★	48.9	33.3	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Processed Gouda cheese spread unseen indicator_group=sensory_quality unseen indicator_type=taste_smell_consistency_change
unseen treatment_type=temperature_storage unseen physical_form=spreadable

EXT_MONTASIO24_MONOMAP1_30 — semi_hard / Montasio

true 60d → pred 78.4d (err +18.4d)

Renoldi et al. (2024), Packaging and shelf-life of semi-hard Montasio cheese · DOI 10.1016/j.fpsl.2024.101363 · 2024

Montasio at 8±1°C, MonoMAP1_80_CO2_20_N2; shelf-life at 30% panelist rancidity threshold.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
69.5	78.4 ★	72.7	68.6	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

- unseen food_matrix=Montasio
- unseen packaging_type=MonoMAP1_80_CO2_20_N2
- unseen indicator_group=sensory_quality
- unseen indicator_type=rancidity_panel_detection
- unseen indicator_unit=%_panelists
- unseen treatment_type=packaging
- unseen application_method=packaging
- unseen physical_form=portioned_slices_250g
- 'matrix_moisture_pct' = 29.0 is outside the training range [34.2, 54.6]
- 'matrix_fat_pct' = 37.0 is outside the training range [20.2, 32.8]
- 'matrix_protein_pct' = 29.0 is outside the training range [13.1, 28.7]
- 'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]
- 'headspace_co2_pct' = 80.0 is outside the training range [0.04, 75]
- 'indicator_threshold' = 30.0 is outside the training range [2, 10]

EXT_MONTASIO24_MONOMAP1_50 — semi_hard / Montasio

true 90d → pred 78.4d (err -11.6d)

Renoldi et al. (2024), Packaging and shelf-life of semi-hard Montasio cheese · DOI 10.1016/j.fpsl.2024.101363 · 2024

Montasio at 8±1°C, MonoMAP1_80_CO2_20_N2; shelf-life at 50% panelist rancidity threshold.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
69.5	78.4 ★	72.7	68.6	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

- unseen food_matrix=Montasio
- unseen packaging_type=MonoMAP1_80_CO2_20_N2
- unseen indicator_group=sensory_quality
- unseen indicator_type=rancidity_panel_detection
- unseen indicator_unit=%_panelists
- unseen treatment_type=packaging
- unseen application_method=packaging
- unseen physical_form=portioned_slices_250g
- 'matrix_moisture_pct' = 29.0 is outside the training range [34.2, 54.6]
- 'matrix_fat_pct' = 37.0 is outside the training range [20.2, 32.8]
- 'matrix_protein_pct' = 29.0 is outside the training range [13.1, 28.7]
- 'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]
- 'headspace_co2_pct' = 80.0 is outside the training range [0.04, 75]
- 'indicator_threshold' = 50.0 is outside the training range [2, 10]

EXT_MONTASIO24_MULTIMAP30 — semi_hard / Montasio

true 105d → pred 78.4d (err -26.6d)

Renoldi et al. (2024), Packaging and shelf-life of semi-hard Montasio cheese · DOI 10.1016/j.fpsl.2024.101363 · 2024

Montasio at 8±1°C, MultiMAP_80_CO2_20_N2; shelf-life at 30% panelist rancidity threshold.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
69.5	78.4 ★	72.7	68.6	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

- unseen food_matrix=Montasio
- unseen packaging_type=MultiMAP_80_CO2_20_N2
- unseen indicator_group=sensory_quality
- unseen indicator_type=rancidity_panel_detection
- unseen indicator_unit=%_panelists
- unseen treatment_type=packaging
- unseen application_method=packaging
- unseen physical_form=portioned_slices_250g
- 'matrix_moisture_pct' = 29.0 is outside the training range [34.2, 54.6]
- 'matrix_fat_pct' = 37.0 is outside the training range [20.2, 32.8]
- 'matrix_protein_pct' = 29.0 is outside the training range [13.1, 28.7]
- 'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]
- 'headspace_co2_pct' = 80.0 is outside the training range [0.04, 75]
- 'indicator_threshold' = 30.0 is outside the training range [2, 10]

EXT_MONTASIO24_MULTIMAP50 — semi_hard / Montasio

true 120d → pred 78.4d (err -41.6d)

Renoldi et al. (2024), Packaging and shelf-life of semi-hard Montasio cheese · DOI 10.1016/j.fpsl.2024.101363 · 2024

Montasio at 8±1°C, MultiMAP_80_CO2_20_N2; shelf-life at 50% panelist rancidity threshold.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
69.5	78.4 ★	72.7	68.6	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Montasio unseen packaging_type=MultiMAP_80_CO2_20_N2 unseen indicator_group=sensory_quality
unseen indicator_type=rancidity_panel_detection unseen indicator_unit=%_panelists unseen treatment_type=packaging unseen application_method=packaging
unseen physical_form=portioned_slices_250g 'matrix_moisture_pct' = 29.0 is outside the training range [34.2, 54.6]
'matrix_fat_pct' = 37.0 is outside the training range [20.2, 32.8] 'matrix_protein_pct' = 29.0 is outside the training range [13.1, 28.7]
'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9] 'headspace_co2_pct' = 80.0 is outside the training range [0.04, 75]
'indicator_threshold' = 50.0 is outside the training range [2, 10]

EXT_PROVOLONE07_CO2_10 — hard / Provolone (aged 6 months)

true 100d → pred 59.5d (err -40.5d)

Favati et al. (2007), Packaging and shelf-life evaluation of portioned Provolone cheese · DOI 10.1016/j.lwt.2005.12.001 · 2007

Six-month-aged Provolone, ~400 g portions, at 8°C in MAP_10_CO2_90_N2.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
63.4	59.5 ★	69.1	73.4	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Provolone (aged 6 months) unseen packaging_type=MAP_10_CO2_90_N2 unseen indicator_group=chemical_quality
unseen indicator_type=free_fatty_acids_C4_plus_C6 unseen indicator_unit=ppm unseen treatment_type=packaging unseen application_method=packaging
unseen physical_form=portion_approx_400g 'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]
'headspace_n2_pct' = 90.0 is outside the training range [0, 79.1] 'indicator_threshold' = 1200.0 is outside the training range [2, 10]

EXT_PROVOLONE07_CO2_100 — hard / Provolone (aged 6 months)

true 175d → pred 64.6d (err -110.4d)

Favati et al. (2007), Packaging and shelf-life evaluation of portioned Provolone cheese · DOI 10.1016/j.lwt.2005.12.001 · 2007

Six-month-aged Provolone, ~400 g portions, at 8°C in MAP_100_CO2.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
65.6	64.6 ★	73.3	60.8	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Provolone (aged 6 months) unseen packaging_type=MAP_100_CO2 unseen indicator_group=chemical_quality
unseen indicator_type=free_fatty_acids_C4_plus_C6 unseen indicator_unit=ppm unseen treatment_type=packaging unseen application_method=packaging
unseen physical_form=portion_approx_400g 'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]
'headspace_co2_pct' = 100.0 is outside the training range [0.04, 75] 'indicator_threshold' = 1200.0 is outside the training range [2, 10]

EXT_PROVOLONE07_CO2_20 — hard / Provolone (aged 6 months)

true 118d → pred 59.5d (err -58.5d)

Favati et al. (2007), Packaging and shelf-life evaluation of portioned Provolone cheese · DOI 10.1016/j.lwt.2005.12.001 · 2007

Six-month-aged Provolone, ~400 g portions, at 8°C in MAP_20_CO2_80_N2.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
64.0	59.5 ★	69.1	73.3	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Provolone (aged 6 months) unseen packaging_type=MAP_20_CO2_80_N2 unseen indicator_group=chemical_quality
unseen indicator_type=free_fatty_acids_C4_plus_C6 unseen indicator_unit=ppm unseen treatment_type=packaging unseen application_method=packaging
unseen physical_form=portion_approx_400g 'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]
'headspace_n2_pct' = 80.0 is outside the training range [0, 79.1] 'indicator_threshold' = 1200.0 is outside the training range [2, 10]

EXT_PROVOLONE07_CO2_30 — hard / Provolone (aged 6 months)

true 280d → pred 59.8d (err -220.2d)

Favati et al. (2007), Packaging and shelf-life evaluation of portioned Provolone cheese · DOI 10.1016/j.lwt.2005.12.001 · 2007

Six-month-aged Provolone, ~400 g portions, at 8°C in MAP_30_CO2_70_N2.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
64.6	59.8 ★	68.9	73.9	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Provolone (aged 6 months) unseen packaging_type=MAP_30_CO2_70_N2 unseen indicator_group=chemical_quality
unseen indicator_type=free_fatty_acids_C4_plus_C6 unseen indicator_unit=ppm unseen treatment_type=packaging unseen application_method=packaging
unseen physical_form=portion_approx_400g 'headspace_oxygen_pct' = 0.0 is outside the training range [0.5, 20.9]
'indicator_threshold' = 1200.0 is outside the training range [2, 10]

EXT_PROVOLONE07_VAC — hard / Provolone (aged 6 months)

true 190d → pred 60.1d (err -129.9d)

Favati et al. (2007), Packaging and shelf-life evaluation of portioned Provolone cheese · DOI 10.1016/j.lwt.2005.12.001 · 2007

Six-month-aged Provolone, ~400 g portions, at 8°C in vacuum.

RANDOM FOREST	LIGHTGBM	XGBOOST	EBM	SUPPORT LEVEL
64.7	60.1 ★	69.3	72.0	unsupported_categorical

Physical-form support: Experimental — No training-support information is available for this combination.

unseen food_matrix=Provolone (aged 6 months) unseen indicator_group=chemical_quality unseen indicator_type=free_fatty_acids_C4_plus_C6
unseen indicator_unit=ppm unseen treatment_type=packaging unseen application_method=packaging unseen physical_form=portion_approx_400g
'indicator_threshold' = 1200.0 is outside the training range [2, 10]

5. Deep dive: hard cheese on real data

All 5 hard-cheese rows come from one paper (Provolone, aged 6 months, five packaging/CO₂ conditions). True values span 100–280d; predictions cluster at 59.5–64.6d — the model is nearly insensitive to what actually distinguishes these five real cases. MAE=111.9d, bias=-111.9d (systematic underprediction), R²=-3.112.

This mirrors a generalization weakness that has recurred across multiple model generations in this project's history for long-aged hard cheese — the V7 retrain and architecture change have not resolved it. It is reported as-is; no correction was applied.

6. Bug found and fixed while building this evaluation

`ModelService.assess_support()` compared every numeric field's raw value against its training range without first checking for `NaN` (as opposed to `None`) — the natural result of `pandas.read_csv` on an empty cell, which several rows in this external file have (e.g. missing `matrix_ph`). A `NaN` always fails a range comparison, so it was being reported as a false "value=nan is outside the training range" extrapolation instead of being treated as missing data; the same applied to categorical `NaNs` being reported as an unseen category "value='nan'". Fixed with a shared missing-value check used in both the categorical and numeric loops, verified directly, and re-confirmed the full 23-case pytest suite (`test_v6_specialist_routing.py`) still passes since this is shared V6/V7 infrastructure.

7. Recommendations

1. Build (or request) a version of this external file using V7's own trained vocabulary for `food_matrix/physical_form/packaging_type/indicator_type`, so accuracy can be assessed independent of vocabulary coverage.
2. Do not remap or fuzzy-match the current file's vocabulary retroactively to make the score look better — a wrong mapping would be worse than an honest "unsupported" flag.
3. Hard cheese needs targeted attention specifically for long-aged profiles (Provolone-like) before any external validation claim is made for that category — consistent with every prior model generation in this project.
4. Semi-hard is the most promising generalizer here (R²=0.560) despite the same vocabulary-mismatch handicap — worth prioritizing if choosing where to invest further validation effort first.

Report generated from `data/raw/CHEESE_SHELF_LIFE_V7_EXTERNAL_TEST_POINT_TARGETS.csv` via `evaluate_v7_external_test.run_evaluation()`, against the live artifacts `v7/specialists`. Full machine-readable output: `v7_external_test_predictions.csv`. No retraining, data modification, or prediction correction occurred in the course of this evaluation or report.